

Axiomatic formal system

Σ_P : set of propositional variables

F_P : set of all formulas

A_P : set of axioms

↳ what is an axiom? We don't prove them
They are tautologies.

R_P : set of inference rules

↳ "modus ponens" - based on implication

MODUS PONENS $\boxed{U, U \rightarrow V \vdash_{mp} V}$ = "from the facts U and $U \rightarrow V$ we can deduce V ".

Deductions hypothesis conclusion

① $U_1, \dots, U_m \vdash V$

V conclusion

↳ is deducible

$(f_1, \dots, f_n)$ = deduction of V from $U_1, \dots, U_m$.

$\begin{cases} u \vdash v : v \text{ logical consequence} \\ u \vdash v : v \text{ syntactical consequence} \end{cases}$

② Theorem = a formula derivable from the axioms and using MODUS PONENS as an inference rule.

- we don't need hypothesis
- we need axioms.

$\vdash v$
 $\vdash u$

Properties of derivability relation

$R \vdash U$ (from a set of formulas we can derive U)

1. monotonicity

$\left. \begin{array}{l} R \vdash U \\ R \subseteq S \end{array} \right\} \Rightarrow S \vdash U$

2. cut

$\left. \begin{array}{l} S \vdash V_j \\ S \cup \{V_1, \dots, V_m\} \vdash U \\ \text{(we add them to the set)} \end{array} \right\} \Rightarrow S \vdash U$

3. transitivity

$\left. \begin{array}{l} S \vdash U \\ \{U\} \vdash V \end{array} \right\} \Rightarrow S \vdash U \wedge V$

... in conclusions

$$4. \text{ conjunction in premises } \left. \begin{array}{l} S \vdash U \\ S \vdash V \end{array} \right\} \Rightarrow S \vdash U \wedge V$$

$$5. \text{ disjunction in premises } \left. \begin{array}{l} S \cup \{U\} \vdash Z \\ S \cup \{V\} \vdash Z \end{array} \right\} \Rightarrow S \cup \{U \vee V\} \vdash Z$$

1. Soundness theorem

a theorem is a tautology

2. Completeness theorem

a tautology is a theorem.

Both:

$$(\vdash U \Leftrightarrow \models U)$$

Consequences:

- 1) non-contradictory (no $\vdash U$ and $\vdash \neg U$ at the same time)
- 2) coherent (not every formula is a ^{theorem} tautology)
- 3) decidable (formula ^{theorem / not} _{tautology / not})

Example ①

$$\begin{array}{ll} \rightarrow H: \text{Harry will go} & \xrightarrow{\text{translate}} H_1: L \wedge \neg G \rightarrow H. \\ L: \text{Lucy will go} & H_2: J \rightarrow L \\ G: \text{George will go} & H_3: J_t \rightarrow J \\ J: \text{John will go} & H_4: \neg G \\ J_t: \text{John in town} & H_5: J_t \\ & C: H. \end{array}$$

Inference rule (it is enough to work w one)

$$\boxed{U, U \rightarrow V \vdash_{mp} V} \text{ MODUS PONENS}$$

$$\left\{ \begin{array}{l} H_5, H_3 \vdash_{mp} J \quad (f6) \\ f6, H_2 \vdash_{mp} L \quad (f7) \end{array} \right.$$

$$\left\{ \begin{array}{l} H_4, \neg 7 \vdash L \wedge \neg G \quad (\neg 8) \\ \neg 8, H_1 \vdash_{mp} H \quad (\neg 9 = C) \end{array} \right.$$

Example ②

$$\neg p \vee q \equiv p \rightarrow q$$

$$p \vee r \equiv \neg p \rightarrow r$$

$$\neg q \vdash r$$

Propositional inference rules

1. addition $\rightarrow$ we can derive a disjunction from $U \rightarrow$ inference rule, theorem
2. simplification:
3. modus ponens: $U, U \rightarrow V \vdash V$
4. modus tollens: $U \rightarrow V \vdash \neg V \rightarrow \neg U$; $\neg V, U \rightarrow V \vdash \neg U$
5. syllogism: $U \rightarrow V, V \rightarrow Z \vdash U \rightarrow Z$; $U, U \rightarrow V, V \rightarrow Z \vdash Z$
6. resolution:

Modus ponens affirming	Modus tollens denying
$\rightarrow$ truth table for understanding	When is $\neg q$ T? When q F. When q F?
$\left. \begin{array}{l} p \text{ T} \\ p \rightarrow q \text{ T} \end{array} \right\} \Rightarrow q \text{ T}$	$\left. \begin{array}{l} q \text{ F} \\ p \rightarrow q \text{ T} \end{array} \right\} \Rightarrow p \text{ F}$
"$\rightarrow$ by affirming the premise (p) affirms the conclusion (q)"	"$\rightarrow$ by denying the conclusion (q) denies the premise (p)"

① if p then q $\left. \begin{array}{l} p \rightarrow q \end{array} \right\}$ sufficiency

② only if p then q $\left. \begin{array}{l} \neg p \rightarrow \neg q \equiv q \rightarrow p \end{array} \right\}$ necessity

	p	q	$p \rightarrow q$
i1	T	T	T
i2	T	F	F
i3	F	T	T
i4	F	F	T

Fallacies

$\hookrightarrow$ mistaken belief based on unsound argument.

1. Fallacy of affirming the conclusion.
2. Fallacy of denying the hypothesis.
3. Fallacy of affirming a disjunct.
4. Fallacy of denying a conjunct.

True: $11 \rightarrow 12$? (with axioms)

$$f_1: \neg U$$

$$f_2: \frac{\neg U}{x} \rightarrow \left(\frac{\neg V}{y} \rightarrow \frac{\neg U}{x} \right) \quad (A_1)$$

$$A_1: x \rightarrow (y \rightarrow x)$$

$$A_3: (x \rightarrow y) \rightarrow (\neg y \rightarrow \neg x)$$

x, y arbitrary formulas.

can I apply modus ponens?

$$f_1, f_2 \vdash_{mp} \neg V \rightarrow \neg U \quad (f_3)$$

$$f_4: (\neg V \rightarrow \neg U) \rightarrow (U \rightarrow V) \quad A_3.$$

$$f_4, f_3 \vdash_{mp} V$$

Theorem of deduction

How to prove that a formula is a theorem

If $U_1, \dots, U_{m-1}, U_m \vdash V$, then $U_1, \dots, U_{m-1} \vdash U_m \rightarrow V$.

Reverse of the theorem:

If $U_1, \dots, U_{m-1} \vdash U_m \rightarrow V$, then $U_1, \dots, U_{m-1}, U_m \vdash V$

Apply the theorem n times

④ Prove that $\vdash (p \rightarrow r) \rightarrow ((p \wedge r \rightarrow q) \rightarrow (p \rightarrow q))$ is a theorem. (without axioms)

SYNTACTIC METHOD:

Step 1. Reverse of the theorem of deduction $\leftarrow$

if $\vdash (p \rightarrow r) \rightarrow ((p \wedge r \rightarrow q) \rightarrow (p \rightarrow q))$ then $p \rightarrow r \vdash ((p \wedge r \rightarrow q) \rightarrow (p \rightarrow q))$

then $p \rightarrow r, p \wedge r \rightarrow q \vdash p \rightarrow q$

then $\underbrace{p \rightarrow r, p \wedge r \rightarrow q, p \vdash q}_{f_1, f_2, f_3} \quad (*)$

Step 2. Prove (*)

$$f_1: p \rightarrow r$$

$$f_2: p \wedge r \rightarrow q$$

$$f_3: p$$

$\hookrightarrow$ Apply modus ponens: $U, U \rightarrow V \vdash_{mp} V$

$$f_1, f_3 \vdash_{mp} r \quad (f_4)$$

$$f_4, f_2 \vdash_{mp} p \wedge r \quad (f_5)$$

$$f_5, f_2 \vdash_{mp} q \quad (\text{Conclusion})$$

premises of the implications
(could be guessed)

how many theorems could we
obtain from here? 6
(mutation of 3)

Step 3: theory of deduction.

if $\begin{matrix} p \rightarrow r, & p \wedge r \rightarrow q, & p \vdash q \\ f_1 & f_2 & f_3 \end{matrix}$ then $p \rightarrow r, p \wedge r \rightarrow q \vdash p \rightarrow q$
 then $p \rightarrow r \vdash (p \wedge r \rightarrow q) \rightarrow (p \rightarrow q)$
 then $\vdash (p \rightarrow r) \rightarrow (p \wedge r \rightarrow q) \rightarrow (p \rightarrow q)$

← permutations to find all the theorems

$\vdash T_2: (p \rightarrow r) \rightarrow p \rightarrow ((p \wedge r \rightarrow q) \rightarrow q)$
 $\vdash T_3: p \rightarrow (p \rightarrow r) \rightarrow ((p \wedge r \rightarrow q) \rightarrow q)$
 $\vdash T_4: p \rightarrow (p \wedge r \rightarrow q) \rightarrow ((p \rightarrow r) \rightarrow q)$
 $\vdash T_5: \dots$
 $\vdash T_6: \dots$

1 2 3
 1 3 2
 2 1 3
 2 3 1
 3 1 2
 3 2 1

5, 6, 7, 8: formula $\rightarrow$ CNF $\rightarrow$ "work as you saw" $\rightarrow$ models in the lectures
 (don't use truth tables)

DIRECT METHOD

Theorem (semantic results) $\rightarrow$ 4th bulletpoint $\rightarrow$ refutation method

Contradiction = inconsistent set.

PROOF BY CONTRADICTION

semantic tableaux

resolution method ??